LOCAFind

A Campus-Based, Location-Enabled Lost and Found Mobile Application

PROJECT CONCEPT & IMPLEMENTATION PLAN

Nueva Ecija University of Science and Technology – Sumacab Main Campus
Mobile Application Technology · Final Case Study
Ionic Angular · Capacitor · Leaflet · PHP + MySQL

1. Project Overview

LOCAFind is a student-driven lost-and-found mobile application that helps the NEUST Sumacab campus community report, discover, and recover misplaced belongings through an interactive, location-based map. Users can report an item they lost or an item they found, attach a photo, give a description, and mark the location. The main screen is a tilted 3D map of the real campus, inspired by the discovery-oriented map of Pokémon GO: each student appears as a cute animal explorer avatar that walks as their phone moves, and lost-item reports appear as mystery markers that only reveal their details when the student walks into the item's search perimeter.

The application makes it easier for students to find out about missing belongings without repeatedly asking around or physically checking with campus security. It supports direct coordination between the person who lost an item and the person who found it, while recognizing that some found items may already have been surrendered to campus security.

2. Problem Statement

Students commonly surrender found belongings to campus security guards. Students who lose items often visit or ask the guards whether an item matching their description—such as an umbrella of a particular color or brand—has been turned in. Guards may need to check the physical storage of recovered belongings, and students have no convenient way to know whether an item has been found before making an inquiry.

Lost-item information may also be shared informally or inconsistently, for example in group chats. A student who loses an item may not know where to look, while another student who finds it may not know how to reach its owner. This creates a gap between reporting a missing item and discovering a relevant found-item report.

LOCAFind addresses this problem by providing a centralized, student-accessible reporting and discovery platform. Its map and location information help users see where items were last seen or found, while photos and descriptions help users identify possible matches.

3. Proposed Solution and Intended Users

The proposed solution is a mobile application with an interactive 3D campus map as its main screen. Students submit lost-item reports with a last-known location and a search perimeter, or found-item reports with a photo, description, and the location where the item was found. Other users discover reports by walking around campus, browse the full list, inspect item details, and contact the reporter to recover an item. Reports go live immediately; there is no admin approval step.

- **Primary users:** NEUST students and other authorized campus community members.
- **Lost-item reporter:** a user who submits details about an item they misplaced.
- **Found-item reporter:** a user who photographs and reports an item they discovered, and records who holds it now.
- **Campus security (outside the app):** keeps surrendered items physically; the app only tells owners where to claim them. Guards do not need to use the app.
- **Optional administrator:** not required for the core prototype; moderation is listed as a future improvement.

4. Core Features

4.1 3D Discovery Map and Mystery Markers

- Show the real NEUST Sumacab campus on a map tilted in 3D, with a sky and horizon. The campus outline comes from OpenStreetMap and everything outside it is dimmed, so the play area stands out.
- Display active lost-item reports as grey "?" mystery orbs, found-item reports as green crates, and campus security posts as purple surrender towers.
- Keep the student's avatar near the bottom centre of the screen; the map follows as they walk (follow camera). Swipe left or right to turn the camera; pinch to zoom.
- Draw each lost item's search perimeter as a circle on the ground, and show an always-visible legend for the marker types.

4.2 How the Search Perimeter Works

Every lost-item report has a **search perimeter**: a circle around the last-known spot that shows the area where the item may be, not an exact point. The reporter chooses how sure they are (exact spot = 30 m, about 50 m, or about 100 m). Each student also has a **100 m**

detection range, shown as a ring around the avatar with ripples spreading from its feet.

State	When	What the student sees	Tapping it shows
Far	More than 100 m away	Faint grey "?" orb with a grey dashed perimeter	Locked card: item silhouette and distance only
Near	Within 100 m, outside the perimeter	Orb glows yellow and bobs faster; perimeter turns yellow; the radar ring becomes a dashed alert; item appears in the Nearby panel	Locked card with distance
In range	Inside the item's perimeter	Phone vibrates, a banner appears, the orb drops in as a red pin, and the perimeter turns red	Full details: photo, description, time, perimeter, reporter contact, "I found this"
Found item	Always	Green crate	Found-item card with custody and "This is mine"
Surrender point	Always	Purple beacon tower	Name, hours, and how to surrender an item
Recovered	Owner closes the report	Removed from the map	Still listed in the reporter's Profile

4.3 Report a Lost Item

- Enter item name, category, description, and an optional photo.
- Pin the last-known location on a small map, starting from the device's current location; tap or drag to adjust.
- Choose the search perimeter (exact spot, about 50 m, or about 100 m).
- Enter a display name and contact method, then submit. The time is recorded automatically.

4.4 Report a Found Item

- Take or select a photo using the device camera, preview it, and confirm or retake it. A photo is required for found items.
- Enter item name, category, and a general description (no identifying details such as the contents of a wallet).
- Confirm or adjust the pin where the item was found.
- Indicate who holds the item now: with the finder, or surrendered to campus security.

4.5 Browse and Search Reports

- List all active lost and found reports, sorted by distance, with search and Lost/Found filters.
- Show basic details (name, category, distance, custody) even when the student is far away, so the app is usable off campus.

4.6 Nearby Panel and Surrender Points

- A NEARBY panel shows silhouettes of missing items within 100 m; tapping it lists each item and its distance in metres.
- The same panel lists the surrender points (Guard House, West Gate guard post, Administration Building) with distances.

4.7 Explorer Avatar and Profile

- Each student picks an original chibi animal explorer (cat, dog, bunny, bear, frog, or fox) and a display name, so no one has to choose a gender. The avatar faces the way it walks: back view when walking away from the camera, front view when coming toward it, and turned to the side otherwise.
- The Profile shows stats (items returned, items surrendered, items recovered, reports made), six achievement badges such as Good Samaritan and Campus Hero, and the student's own reports with status buttons.

4.8 Bird's-Eye View

- Tapping the compass offers "Bird's-eye view" and "Face north". The bird's-eye view flattens the camera, zooms out to the whole campus, shows building labels from OpenStreetMap, and lets the student drag to look around, with "Find me" and "Back" buttons.

4.9 Recovery Status

- The reporter updates a report from their Profile: mark a found item "Surrendered to Campus Security", or close a report as "Returned" / "I got it back".
- Statuses: Active (Lost or Found), Surrendered to Campus Security, and Recovered/Closed. Closed reports leave the map.
- A report posted by mistake can be deleted from the Profile after a confirmation; it is removed from the database for everyone.

5. User Interface and Navigation

The app uses a labelled bottom navigation bar on every screen: **Map · Browse · Report · Profile**. Report opens a choice of Report Lost Item or Report Found Item. The Map screen adds a small game-style HUD around the 3D view:

Position	Element	What it does
Bottom left	Explorer portrait and name	Opens the explorer card: choose an animal and display name, Save, or go to the Profile
Bottom right	Compass	Options: Bird's-eye view or Face north; in the bird's-eye view it returns to the explorer view
Bottom right	NEARBY panel (below the compass)	Silhouettes of nearby missing items; expands to the list of items and surrender points
Top right	Legend	Always visible: Missing, Near you, Unlocked, Found item, Surrender
Top left	Notices	Offline mode, location problems, or Demo mode

6. User Workflow

Workflow A — Reporting a Lost Item

1. The student taps Report, then Report Lost Item.
2. The student enters the item's name, category, and description, and optionally adds a photo.
3. The student pins the last-known location; geolocation suggests a starting point and the pin can be dragged.
4. The student chooses the search perimeter (exact spot, about 50 m, or about 100 m).
5. The student submits the report. A grey "?" orb and its perimeter appear on every user's map.

Workflow B — Reporting a Found Item

1. The student taps Report, then Report Found Item.
2. The student takes a photo, previews it, and confirms or retakes it.
3. The student enters a category and a general description.
4. The app captures the location where the item was found, and the student may adjust the pin.
5. The student records whether the item is still with them or has been surrendered to campus security.
6. The report is submitted and a green crate appears on the map and in Browse.

Workflow C — Discovering and Recovering an Item

1. The student walks around campus with the Map open, or searches the Browse list.
2. When the radar shows an item near them, the student opens the Nearby panel and walks toward it.
3. Inside the perimeter, the phone vibrates and the orb becomes a pin; tapping it shows the full details.
4. The student taps "I found this" (on a missing item) or "This is mine" (on a found item). A message shows the reporter's name and contact, or tells the owner to claim the item from Campus Security if it was surrendered.
5. The two students contact each other outside the app (for example on Messenger) and arrange the return, or the owner goes to the security post and describes the item to the guard.

6. Once the item is back with its owner, the student who posted the report closes it from their Profile, and it leaves the map. Workflow E explains each step in detail.

Workflow D — Surrendering an Item to Campus Security

1. The student finds the nearest surrender point in the Nearby panel or on the map.
2. The student hands the item to the guard; the guard does not need to use the app.
3. In the Profile, the student marks the report "Surrendered to Campus Security".
4. The owner sees on the report card where to claim the item.

Workflow E — What Each Button Does

The detail card is where a finder and an owner meet. The table below lists exactly what the app does for each button, as implemented in the prototype.

The student taps	Where	What the app does	What is saved
A missing-item marker (far or near)	Map	Shows a locked card with the category silhouette and the distance: "Get closer to see details."	Nothing
A missing-item marker (inside its perimeter)	Map	Shows the full card: photo, name, description, time reported, distance, and perimeter.	Nothing
"I found this"	Missing-item card	Opens a message: "Found this item? Reach [reporter] at [contact] to arrange the return." The student contacts the owner outside the app.	Nothing; the owner is not notified by the app
A found-item crate	Map	Always shows the full card, including who holds the item now: with the finder or surrendered to Campus Security.	Nothing
"This is mine"	Found-item card	Opens a message with the finder's name and contact. If the item was surrendered, it tells the owner to claim it at the surrender point and describe it to the guard.	Nothing; the finder is not notified by the app
"Surrendered"	Profile (finder's own found report)	Marks the item as surrendered to Campus Security right away, after the finder hands it to the guard.	PUT: custody = security
"Returned"	Profile (finder's own found report)	Asks "Returned to the owner?", then closes the report and removes it from the map.	PUT: custody = returned, status = recovered
"I got it back"	Profile (owner's own lost report)	Asks "Got it back?", then closes the report and removes it from the map.	PUT: status = recovered
"Delete"	Profile (own report)	Warns that the report is removed for everyone and suggests closing it instead if the item was really returned.	DELETE: row removed

How the two people connect. The app does not include chat. It shows each reporter's contact (for example "Messenger: Juan Dela Cruz") so the other student can reach them directly. For a missing item, the contact is shown only after the student walks inside its perimeter, so the details are seen by people who are actually near where the item was lost.

How a case is closed. Only the student who posted a report can change it, and each report is closed on its own. If both a lost report and a found report exist for the same umbrella, the owner closes the lost report with "I got it back" and the finder closes the found report with "Returned". The app does not link the two reports or wait for both people to confirm; this keeps the prototype simple, and it matches how a return happens in person. A report that is never closed stays on the map until its owner closes or deletes it.

Proving ownership. Found-item descriptions stay general, so a claimant has to describe the item in person to the finder or to the guard before receiving it. The app does not verify claims.

> **Testing without the server or off campus.** If the phone cannot reach the XAMPP laptop, the app runs in offline mode with five sample reports on the NEUST map. Choose "Continue as guest", then turn on Profile → Demo mode and tap the map to walk. Missing items unlock when the avatar enters their perimeter, and both "I found this" and "This is mine" work with the sample contacts. Reports posted offline are saved on that phone only.

7. System Roles and Administration

The core system is student-driven. There is no required admin approval before lost or found reports appear, and guards are not required to register items or manage the application, because their regular smartphone use and participation cannot be assumed.

Students create an account with their student number, a display name, and a password (stored only as a bcrypt hash). Logging in gives the phone a token that is sent with every change; posting a report needs a login, and only the student who posted a report can update or delete it. Because reports, achievements, and the explorer's look belong to the account, they follow the student to any phone. Guests can explore the map and browse reports but cannot post. A full admin dashboard is not essential to the project's central purpose; administration, moderation, and claim verification are acknowledged as limitations and future improvements. The app does not replace campus security or physically manage recovered belongings.

8. Camera, Geolocation, Map, and Data Implementation

Layer	Choice	Why
UI framework	Ionic + Angular (NgModule pages)	Required by the course; tabs, action sheets, alerts, and toasts cover the interface
Native bridge	Capacitor (Android)	Turns the web app into an installable Android app (APK)
Camera	@capacitor/camera	Take or choose the found-item photo, with preview, confirm, and retake
Location	@capacitor/geolocation	Live position for the radar and avatar; starting point for report pins
Haptics	@capacitor/haptics	Vibration when the student enters a missing item's perimeter
Accounts	PHP auth.php + password_hash + login tokens	Sign up and log in with a student number; the token is sent as "Authorization: Bearer" by an Angular HTTP interceptor
Map	Leaflet + OpenStreetMap tiles	Free, no API key; draws the ground, campus outline, and perimeters
Data	Angular HttpClient + PHP + MySQL (XAMPP)	Same pattern as the class students.php lesson: full CRUD (GET, POST, PUT, DELETE) with JSON

Camera

The Capacitor Camera plugin takes or selects an image for a found-item report. The photo is compressed (quality 50, at most 800 px wide) and shown in a preview with Confirm and Retake actions before submission. A lost-item report may include an optional photo.

Geolocation

Capacitor Geolocation watches the device's position while the Map or Report screen is open. The avatar glides between position updates, and distances to every report are recalculated with the haversine formula to decide each marker's state (Far, Near, In range). Users can always place or adjust a pin manually, because GPS may be inaccurate indoors.

Interactive 3D Map

Leaflet draws the ground (tiles, campus outline, and perimeters) on a map element tilted with a CSS perspective transform. Items, towers, and the avatar are upright elements in a flat layer on top, placed every frame by projecting their map position through the same transform and scaled by distance. Taps on the ground are converted back to map coordinates by inverting the projection. The map clearly communicates that a marker is a user-reported location, not live tracking of the physical item.

Data / API

The backend is one procedural PHP file (api/reports.php) with PDO and a MySQL reports table, following the instructor's students.php example: JSON responses shaped { success, data }, a CORS allow-list, and validation with 400, 404, 405, and 422 status codes. The app covers full CRUD: GET to load reports, POST to submit a report, PUT to change a report's status or custody, and DELETE to remove a report posted by mistake. Closing a report keeps it as a record, while deleting removes it for good. If the server cannot be reached, the app switches to offline mode: it shows sample reports and saves new reports on the phone only.

9. Data Fields

Field	Type	Used for	Notes
id	number (auto)	Lost, Found	Generated by MySQL on insert
user_id	number	Lost, Found	Account that posted the report (users table: student number, display name, password hash, contact, avatar)
type	lost / found	Lost, Found	Chooses the marker style
name, category	text	Lost, Found	Category picks the silhouette (ID, bag, umbrella, gadget, tumbler, clothing, keys, other)
description	text (max 240)	Lost, Found	General only; no identifying details
photo	base64 image	Optional / required	Optional for lost, required for found
lat, lng	decimal	Lost, Found	Last-known or found location
radius	metres	Lost	Search perimeter (30 m minimum)
reporter_name, contact	text	Lost, Found	Contact shown only to students inside the perimeter
custody	with_finder / security / returned	Found	Who holds the item now
status	active / recovered	Lost, Found	Recovered reports leave the map
created_at	timestamp	Lost, Found	Set by the database

10. Scope and Features to Defer

The case study asks for a focused, small-scale working solution. The group prioritizes the report and map workflow, the required device capabilities, and the data operations, and defers features that are not necessary to demonstrate the idea.

Implemented core scope

- 3D discovery map of the real campus with mystery markers, perimeters, and the walking avatar
- Report lost item and report found item, with camera, geolocation, and manual pin adjustment
- Proximity reveal (Far, Near, In range), Nearby panel, surrender points, and vibration alert
- Browse with search and filters; bird's-eye view with building labels
- Full CRUD (GET, POST, PUT, DELETE) through HttpClient to the PHP + MySQL API, with an offline fallback
- Profile with avatar, display name, achievements, and report status updates
- Accounts: sign up and log in with a student number and password; guests can browse; only the owner can update or delete a report
- Demo mode for testing away from campus (tap the map to walk)

Defer to future improvements

- Real-time chat or push notifications
- AI image recognition or automatic visual matching
- Full admin dashboard, multi-role permissions, and moderation
- Guard accounts and a guard-managed item inventory
- Password reset, email verification, and checking student numbers against NEUST records
- Formal claim approval or identity verification
- Shared leaderboards across all students

11. Limitations and Safety Considerations

- The app depends on user-submitted reports; it cannot guarantee that every lost or found item will be reported.
- The app does not track an item in real time. A marker shows where an item was reported, not where it is now.
- GPS may be inaccurate indoors or in dense campus areas, so users can adjust pins manually and perimeters have a 30 m minimum.
- The XAMPP server runs on the group's laptop: the phone must be on the same Wi-Fi, and the laptop's IP address may change. When the server cannot be reached, the app works in offline mode with sample data.
- Off campus, every marker stays locked; the Browse tab and Demo mode let the instructor test the app anywhere.
- Accounts are not checked against official NEUST records, there is no password reset yet, and the API runs over plain HTTP on the local network (passwords are hashed in the database, but HTTPS hosting is a future improvement).
- There is no in-app messaging: tapping "I found this" or "This is mine" shows the other person's contact but does not notify them, and the app does not record that anyone tapped it.
- Lost and found reports for the same item are not linked, and a return is not confirmed by both people; each reporter closes their own report.
- Public item reports should avoid sensitive identifying details that could enable false claims; owners prove ownership in person or with the guard.
- A user who loses their phone may need another device; the app is not a replacement for Find My Device.
- Watching the location uses battery, so it runs only while the Map or Report screen is open.

12. Future Improvements

- Make the contact tappable, so "I found this" opens Messenger, a call, or an SMS directly.
- Save a claim when someone taps "I found this" or "This is mine", and show it in the reporter's Profile, so the reporter knows someone responded.
- Notifications when a found report appears near a lost item's last-known location.
- Matching suggestions based on item category, description, time, and distance.
- Password reset and verification of student numbers with the registrar.
- Optional admin moderation for inappropriate, duplicate, or fraudulent reports.
- Optional campus-security integration if guards and campus management agree to use the system.
- Hosting the API online with HTTPS so the app works on any network.

13. Summary

LOCAFind is a student-driven, campus-based lost-and-found application built around a game-like 3D map of the NEUST Sumacab campus. Students report missing belongings with a last-known location and a search perimeter, and report found belongings with a photo, description, and found location. Other students walk around campus as a cute animal explorer; items stay hidden as faint mystery orbs until the student steps inside an item's perimeter, when the phone vibrates and the full details unlock. Campus security remains part of the existing physical process when an item is surrendered, but guards do not need to operate the app. The project focuses on the core requirements of the Mobile Application Technology case study—Camera, Geolocation, an interactive map, and HttpClient data operations with a PHP and MySQL backend—while keeping the scope small enough to finish, test, and explain.